

Identification of Important Nodes in Multilayer Heterogeneous Networks Incorporating Multirelational Information

Liangtian Wan^{ID}, *Member, IEEE*, Mingyue Zhang, Xiaona Li, Lu Sun^{ID},
Xianpeng Wang^{ID}, *Member, IEEE*, and Kaihui Liu^{ID}, *Member, IEEE*

Abstract—Centrality is an effective method to identify important nodes in complex networks, but it is still a challenge to find influential nodes by making full use of multiple relationships and global network topological features in complex networks. To address these problems, this article proposes an importance identification method for multilayer heterogeneous network node by incorporating multirelational information (MLC). This method studies the relational characteristics of heterogeneous nodes in detail and divides the heterogeneous nodes into different layers according to the node types, which can be further divided into core and auxiliary layers. The importance of the auxiliary layer is quantified by designing the interlayer influence and determining the interlayer influence weights of different connectivity influences; the centrality score of heterogeneous nodes under multiconnectivity relationships is fused using the transmission characteristics of internode relationships in the auxiliary layer, which in turn measures the importance of nodes in the core layer. To evaluate the proposed algorithm, we conduct experiments on five real multilayer heterogeneous networks of different sizes. The results show that MLC can make full use of different types of internode association relationship information, effectively fuse network structure information such as the neighbor weights of core and auxiliary layer nodes, and outperform the existing techniques in identifying important nodes.

Index Terms—Centrality, important nodes, interlayer influence, multilayer heterogeneous networks.

Manuscript received 6 January 2022; revised 24 February 2022 and 14 March 2022; accepted 15 March 2022. Date of publication 13 April 2022; date of current version 1 December 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62101088 and Grant 61801076, in part by the Radar Signal Processing National Defense Science and Technology Key Laboratory Fund under Grant 6142401200101, in part by the Fundamental Research Funds for the Central Universities under Grant DUT20JC29, and in part by the Dalian High-level Talent Innovation Support Plan under Grant 2019RQ024. (*Corresponding author: Lu Sun.*)

Liangtian Wan, Mingyue Zhang, and Xiaona Li are with the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, School of Software, Dalian University of Technology, Dalian 116620, China (e-mail: wanliangtian@dlut.edu.cn; zhangmy@mail.dlut.edu.cn; lixiaona@mail.dlut.edu.cn).

Lu Sun is with the Department of Communication Engineering, Institute of Information Science Technology, Dalian Maritime University, Dalian 116026, China (e-mail: sunlu@dlmu.edu.cn).

Xianpeng Wang is with the State Key Laboratory of Marine Resource Utilization in South China Sea and the School of Information and Communication Engineering, Hainan University, Haikou 570228, China (e-mail: wxpeng2016@hainanu.edu.cn).

Kaihui Liu is with the School of Electrical Engineering and Intelligentization, Dongguan University of Technology, and National Key Laboratory of Communications, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: kaihui@outlook.com).

Digital Object Identifier 10.1109/TCSS.2022.3161305

I. INTRODUCTION

IDENTIFYING the importance of nodes, as one of the hot issues in complex network analysis, relies on a theoretical framework mostly oriented to single-layer network structures [1]–[3]. As a simplified model of complex systems, single-layer networks ignore the contribution of multiple types of relational interactions in complex systems. However, real-world data relationships are complex and diverse, and the constituted networks are more often composed of numerous entities, which contain multiple types of nodes and relationships. Also, different types of nodes and relationships between nodes reveal different semantic information. For example, in the academic network [4], in addition to the citation relationship among documents, it also includes the writing relationship between scholars and documents, and the publication relationship between documents and journals, and so on. Such interacting relationships give rise to a variety of associative structures in time and space, and they also exhibit the multidimensional characteristics of complex systems in time and space as well. The single-layer network structures developed so far are not sufficient to accurately portray the interrelated and interdependent entity relationships in the real world [5], [6]. Obviously, the abstraction of complex systems into the form of single-layer networks alone cannot further satisfy the needs of studying complex systems.

Currently, the study of multilayer networks has gradually become a challenging research hotspot in the field of complex network analysis. Compared with a single-layer network structure, multilayer networks break the limitation of homogeneity of node and edge relationships in single-layer network. Multilayer networks pay more attention to heterogeneity in complex systems (multiple types of nodes and their edge relationships, including intralayer and interlayer edges) [7], [8]. Multilayer networks have become a common model for portraying complex systems composed of various types of relationships in the real world. In general, multilayer networks will appear in the form of interacting relationships of several networks. For example, in a social relationship system, different types of social relationships (colleagues, friends, relatives, and so on) can be reduced to different network layers. Thus, social relationships can be represented as business partners, friendships, relatives, and so on. In a

literature network [4], literature may be linked to each other at multiple levels (keywords, references, research areas, authors, and so on), and thus, these relationships can be used to analyze the differences and associations arising from different types of interactions. In an airline network system [9], since different airlines operate different flight routes between multiple identical airports, a comprehensive multilayer network framework can be used to study the ability of airline network systems to deal with unexpected disasters. The network of web pages in Wikipedia [10] can also be abstracted as a multilayer network. Due to the existence of page-hopping relationships between terms (or words) contained in web pages of different topics, the potential influences or connections that may exist in the interaction of web pages of different topics can be studied.

In network science, node centrality [11]–[14], as a research topic and hotspot in complex network analysis, is of great significance in revealing the relationship between network structure and function. In the field of social network analysis, more complete and rich metrics and methods have been established for measuring the centrality of single-layer network nodes [15]. Its typical methods are divided into two categories, which are metrics based on local attributes of nodes and global attributes of nodes. A typical and representative metrics based on the local attributes of nodes is degree centrality (DC) [16], which reflects the direct influence of the current node in the whole network. In addition, since node importance has an important relationship with its neighboring nodes, metrics based on global attributes of nodes seem to be more suitable for measuring node importance, such as betweenness centrality (BC) [17], closeness centrality (CC) [18], eigenvector centrality (EC) [19], and PageRank [20]. However, in the field of multilayer network, the existing work on multilayer networks has focused on basic statistical characteristics [21], network topology [22], [23], propagation spreading [24]–[26], association structure [27], [28], and other fundamental issues, and a series of rich and influential theoretical results has been obtained [29]–[31]. However, there is still little study that addresses the importance of nodes in multilayer networks, and most of them are still expressed by constructing multilayer homogeneous networks and aggregating them by some rules [8], [22]. While this modeling method reduces the difficulty of subsequent studies, it loses the structural information specific to multilayer networks and the heterogeneous information (the interaction relationships among different types of nodes) that exists in complex systems. It can be seen that existing studies still face the following challenges in the problem of identifying important nodes: how to accurately portray multilayer networks while preserving more of the rich heterogeneous information in the networks and how to extend the existing methods for identifying the importance of nodes on a single-layer network to multilayer networks.

To address these challenges, this article addresses the importance problem of node in multilayer networks and proposes a multilayer heterogeneous network node importance identification method incorporating multirelational information (MLC). First, focusing on the multitype relational data generated in complex systems, we represent and analyze them by constructing a multilayer heterogeneous network model and applying a

multilayer heterogeneous network framework based on tensor algebra. Second, we calculate the centrality of nodes at each layer by drawing on K-shell [32] and betweenness centrality [17] and then quantify the neighbor weights of multilayer heterogeneous networks. Finally, by fusing multirelationship information and introducing the attribute characteristics and transmission properties of nodes, we obtain the node importance ranking for multilayer heterogeneous networks. In the experimental part, we compare and analyze five real multilayer heterogeneous network datasets, and the results show that the proposed method in this article effectively improves the accuracy of node importance measurement.

The method proposed in this article not only enriches the node importance measures for multilayer heterogeneous networks but also extends the practical application scenarios of K-shell methods and betweenness centrality. Its main contributions include three aspects.

- 1) A multilayer heterogeneous network node importance identification method incorporating MLC is proposed, which effectively solves the problem of identifying important nodes on multilayer heterogeneous networks. It can achieve this effect through constructing a multilayer heterogeneous network model and measuring the importance of nodes using multirelationship coupled information and transmission mechanisms.
- 2) We introduce a new interlayer influence function, which considers the contribution of interlayer interactions and weighted aggregated interlayer neighborhood features to couple MLC.
- 3) We propose a new intralayer node centrality, which uses the K-shell method and the betweenness centrality to find key nodes in the layer network.

II. RELATED WORK

Due to the introduction of multidimensional network layers, multilayer network models bring new challenges to the portrayal of basic statistical properties, association structures, and dynamic behaviors of networks [33]. Initially, the study of multilayer networks simply extended the concept of single-layer networks, such as multilayer network aggregation representation [8], which is a multilayer structure that compresses a multilayer network into a single-layer network without considering the interactions between the layers (such a single-layer network is called an aggregated network). Alternatively, with the help of theoretical methods such as matrix decomposition and multisubnet synthesis, existing networks are temporally partitioned or snapshot, and then, each network is studied individually before synthesis [34]. This simplified modeling approach loses information about the specific structure contained in the original data and about the associations between different types of relationships, which is not conducive to conducting subsequent studies.

In the field of computer science [6], [35], researchers have studied different types of multilayer networks and their dynamic processes using tensor decomposition methods [36]–[38], and multiway data analysis [39]. Representation of multilayer networks in this way not only directly reflects

the association between different layers but also preserves the detailed information of the network. Lei *et al.* [40] incorporated interlayer information into the process of network aggregation and addressed the heterogeneity of layer interactions. Tejedor *et al.* [41] focused on the effect of the degree of interlayer coupling and asymmetry on the diffusion process in multilayer directed networks, and the results show that moderately coupled networks will have higher propagation capacity than fully coupled networks. To describe the heterogeneity existing in complex systems more comprehensively and completely, each class of nodes and their interactions can correspond to a network layer, and a multilayer heterogeneous network model can be established to carry out related research.

Quantifying node centrality [14], [42]–[44], as a research hotspot in complex network analysis, is important for studying the structure and function of networks. Centrality measurement of multilayer networks can make full use of the rich heterogeneous information and comprehensive multilayer network structure information in the network to solve the identification problem of important nodes. Multilayer networks focus more on the interactions between different network layers, including layer overlap, correlation, and centrality. Halu *et al.* [45] and Pedroche *et al.* [46] defined the PageRank of multilayer networks based on the interactions between layers. Pedroche *et al.* [47] considered the overlap of multiplexed network layers, treated the PageRank of multiplexed networks as a smooth distribution, and accurately estimated the PageRank of this type of network by some constraints. Some researchers have focused more on how interlayer connectivity affects the node centrality of multilayer networks. Wu *et al.* [48] used a tensor framework to study the eigenvector centrality of multilayer networks and proved the uniqueness of the existence of eigenvector centrality of multilayer networks for a given form of interlayer influence. Chen and Lu [49] compressed multilayer networks into aggregated networks, quantified the importance of edges, and then distributed edge centrality to each layer according to some strategy. As a result, they finally define layer centrality as the sum of edge centrality in the layer. Rahmede *et al.* [50] fused layer and node information to determine layer centrality and node centrality, providing new ideas for the study of centrality in multilayer networks. Most of the above methods construct multilayer homogeneous networks, quantifying the importance of nodes through interlayer connections and intralayer topological information. Although the multilayer homogeneous network centrality method retains rich node characteristics and neighborhood structure, it ignores the deep mining of multitype relationship information in the network structure and cannot be directly applied to multilayer heterogeneous networks with complex structures.

The method proposed in this article makes full use of network information, such as node characteristics, relationship characteristics, and neighborhood structure in multilayer heterogeneous networks. Also, we effectively solve the node importance problem of multilayer heterogeneous networks. We consider not only the most local information of the nodes but also the influence of the environment around the nodes so that it can better reflect the topological characteristics of the nodes. In addition, in order to balance the complexity

of global centrality, we add K-shell in our method to strip the nodes to different shells layer by layer. The method can deeply explore the characteristics of multiple types of relationships in heterogeneous networks. We are by using the transmission characteristics of relationships between nodes and the influence of design layers to discover the influence of different connection relationships on node centrality, and the fusion of heterogeneous node centrality scores improves the identification of important nodes.

III. FORMULATION OF MLC

A number of centrality methods have been proposed around the study of identifying important nodes in networks. Different methods consider different structural information in the network and have their advantages and disadvantages as well. Since most previous methods have focused on single-layer network structures, they cannot adequately fuse multirelational features for comprehensive analysis. In this article, we propose a multilayer heterogeneous network node importance identification method incorporating MLC, which can find nodes at the core of the network with better importance and control. The method pays more attention to the heterogeneity in complex systems, which includes different types of nodes and the interaction patterns between nodes belonging to different network layers.

A. Basic Model and Tensor Representation of Multilayer Heterogeneous Networks

In this section, we introduce the basic model and tensor representation of multilayer heterogeneous networks. We use a tensor algebra-based multilayer network representation framework to model and analyze networks [36], [37], [48], [51]. The advantage of this framework is that it is scalable and allows the construction of many different types of networks, such as multinetworks, time-series networks, and interdependent networks.

1) *Basic Model of Multilayer Heterogeneous Networks:* In general, a heterogeneous network can be represented as $G_R = (V, E_R)$, where $V = \{V_t\}_{t=1}^T$ is the set of nodes containing T types, N is the number of vertices in the network, and $E_R = \{E_R^{[1]}, E_R^{[2]}, \dots, E_R^{[M]}\}$ is the set of edges, with $E_R^{[M]}$ the set of edges of the M th class of relations.

For multilayer heterogeneous networks, they can be represented hierarchically as two parts: intralayer networks and interlayer networks, with each layer corresponding to a type of relationship.

Definition 1: The intralayer network is denoted by $G_R^{[a]} = (V^{[a]}, E_R^{[a]})$, and the set of connected edges within the layer can be written as

$$E_R^{[a]} = U_a \left\{ \left(V_i^{[a]}, V_j^{[a]} \right) \mid i, j \in \{1, 2, \dots\} \right\}. \quad (1)$$

Definition 2: The interlayer network is denoted by $G_R^{[a,\beta]} = (V^{[a]}, V^{[\beta]}, E_R^{[a,\beta]})$ and the set of interlayer connected edges can be written as

$$E_R^{[a,\beta]} = U_{a,\beta} \left\{ \left(V_i^{[a]}, V_j^{[\beta]} \right) \mid i, j \in \{1, 2, \dots\} \right. \\ \left. a, \beta \in \{1, 2, \dots, M\} \right\}. \quad (2)$$

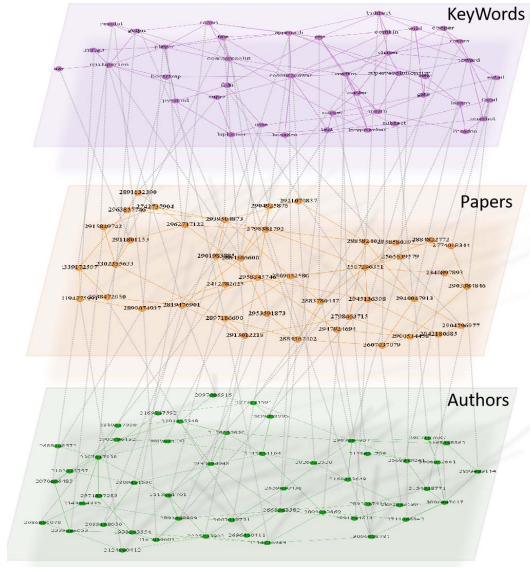


Fig. 1. Heterogeneous network structure of undirected three layers.

A multilayer heterogeneous network can be constructed from Definitions 1 and 2. Fig. 1 shows a schematic of an undirected multilayer heterogeneous network with three layers.

2) *Tensor Representation of Multilayer Heterogeneous Networks*: Multilayer heterogeneous networks can be modeled using the network models mentioned above. This section focuses on the tensor algebra-based framework for multilayer heterogeneous networks. The tensor representation of multilayer heterogeneous networks not only can directly derive the correspondence among different layers but also does not lose the detailed information of the network.

For multilayer heterogeneous networks, each class of relations is represented by a separate layer k ($k = 1, 2, \dots, M$). The second-order tensor $A_{\beta k}^{\alpha}$ is used to represent the intralayer adjacency tensor of the k th layer. In order to represent information containing multiple-layer relations, the second-order interlayer adjacency tensor $C_{\beta k}^{ak}$, which represents the connection from layer h node $V_i^{[h]}$ to layer k node $V_j^{[k]}$, is introduced. In particular, the interlayer adjacency tensor of the same layer k is equal to the intralayer adjacency tensor of that layer, i.e., $C_{\beta k}^{ak} = A_{\beta k}^{\alpha}$.

To facilitate the subsequent study of the centrality of multilayer heterogeneous networks, the intralayer and interlayer adjacency tensor can be calculated and analyzed using a matrix representation. In fact, the adjacency matrices corresponding to the intralayer and interlayer networks can be easily obtained according to Definitions 1 and 2.

Definition 3 (Intralayer Adjacency Matrix): The intralayer connections corresponding to the class α relations in a multilayer heterogeneous network can be represented by an adjacency matrix, denoted as $A^{[\alpha]} \in R^{N \times N}$, and the intralayer adjacency matrix is defined as

$$A = \{A^{[1]}, A^{[2]} \dots A^{[M]}\}. \quad (3)$$

Definition 4 (Interlayer Adjacency Matrix): The interlayer connections corresponding to a multilayer heterogeneous network can be represented by the interlayer network adjacency matrix, denoted as $O^{[\alpha, \beta]} \in R^{N \times N}$, and the interlayer adjacency matrix is defined as

matrix, denoted as $O^{[\alpha, \beta]} \in R^{N \times N}$, and the interlayer adjacency matrix is defined as

$$O = \{0^{[1,2]}, 0^{[1,3]} \dots 0^{[\alpha, \beta]} \mid \alpha \neq \beta\}. \quad (4)$$

According to Definitions 3 and 4, the general form of a multilayer heterogeneous network can be defined as $G = (A, O) = \{A^{[1]}, A^{[2]} \dots A^{[M]}, 0^{[1,2]}, 0^{[1,3]} \dots 0^{[\alpha, \beta]}\} (\alpha, \beta \in \{1, 2, \dots, M\})$. Based on the superadjacency matrix [7], [52], G can be expressed as

$$G = \begin{pmatrix} A^{[1]} & O^{[1,2]} & \dots & O^{[1,M]} \\ O^{[2,1]} & A^{[2]} & \dots & O^{[2,M]} \\ \vdots & \vdots & \ddots & \vdots \\ O^{[M,1]} & O^{[M,2]} & \dots & A^{[M]} \end{pmatrix}. \quad (5)$$

B. Multilayer Heterogeneous Network Node Importance Identification Method Incorporating Multirelational Information

Based on the ideas of K-shell [32] and betweenness centrality [17], we generalize these methods to multilayer heterogeneous networks. A multilayer heterogeneous network important node identification method based on multirelationship fusion (MLC) is proposed. MLC considers the node itself, its neighboring nodes' centrality, and prestige along with the multilayer network structure properties under the mode of action of multiple types of relationships. MLC includes the following three definitions.

Definition 5 (Interlayer Influence): The interactions between nodes should not only consider the interactions between neighboring nodes in the same layer but also combine the interactions between heterogeneous nodes in different layers. Therefore, the interlayer influence is set to be quantified based on the interlayer connectivity of the network. It is considered that the more the number of common nodes in the network where two neighboring nodes are connected to other layers, the greater the influence effect of that layer on the nodes, and the higher the weight of the interlayer influence. The interlayer influence ω_{β}^{α} in a multilayer heterogeneous network represents the influence weight from layer α to layer β in a multilayer network G , which can be defined as

$$\omega_{\beta}^{\alpha} = \frac{\sum_{(V_i^{[\beta]}, V_u^{[\alpha]}) \in E^{[\alpha, \beta]}, (V_j^{[\beta]}, V_v^{[\alpha]}) \in E^{[\alpha, \beta]}} \frac{n_{v_i^{[\beta]}}(v_u^{[\alpha]}) \cap n_{v_j^{[\beta]}}(v_v^{[\alpha]})}{n_{v_i^{[\beta]}}(v_u^{[\alpha]}) \cup n_{v_j^{[\beta]}}(v_v^{[\alpha]})}}{\sum_{(v_i^{[\beta]}, v_j^{[\beta]}) \in E^{[\beta]}} n_{v_i^{[\beta]}}(v_j^{[\beta]})} \quad (6)$$

where $n_{v_i^{[\beta]}}(v_j^{[\beta]})$ denotes the number of connected edges from node $v_i^{[\beta]}$ to node $v_j^{[\beta]}$ in layer β , $n_{v_i^{[\beta]}}(v_u^{[\alpha]})$ denotes the number of connected edges from node $v_i^{[\beta]}$ in layer β to node $v_u^{[\alpha]}$ in layer α , and $n_{v_j^{[\beta]}}(v_v^{[\alpha]})$ denotes the number of connected edges from node $v_j^{[\beta]}$ in layer β to node $v_v^{[\alpha]}$ in layer α .

Definition 6 (Intralayer Node Importance): In fact, the higher the degree of the node and the more neighbors the node has, so such nodes are more likely to be more important. On the other hand, nodes in key topological locations of the network should also have higher influence. In this case,

Algorithm 1 MLC

Input: $G_R = (V, E_R)$;
 Network G which contains N nodes and E_R edges;
 1: Initialize the information
 2: **for** each layer $\alpha \in \{1, 2 \dots M\}$ **do**
 3: compute ω_β^α using Eq.(6)
 4: **end for**
 5: **for** each layer $\alpha \in \{1, 2 \dots M\}$ **do**
 6: **for** each node $v_x^{[\alpha]} \in \alpha$ **do**
 7: compute $ILC_{v_x^{[\alpha]}}^\alpha$ using Eq.(7)
 8: **end for**
 9: **end for**
 10: **for** each node $v \in V$ **do**
 11: compute $ILC_{v_x^{[\alpha]}}^\alpha$ using Eq.(8)
 12: **end for**

we jointly use the K-shell method and betweenness centrality to uniquely determine the importance of nodes within a layer. The importance of node $v_x^{[\alpha]}$ within layer α is denoted by $ILC_{v_x^{[\alpha]}}^\alpha$ and can be defined as

$$ILC_{v_x^{[\alpha]}}^\alpha = \phi_{v_x^{[\alpha]}}^\alpha + \sum_{v_s^{[\alpha]} \neq v_x^{[\alpha]}, v_t^{[\alpha]} \neq v_x^{[\alpha]}} \frac{\sigma_{v_s^{[\alpha]} v_t^{[\alpha]}}^\alpha(v_x^{[\alpha]})}{\sigma_{v_s^{[\alpha]} v_t^{[\alpha]}}^\alpha} \quad (7)$$

where $\phi_{v_x^{[\alpha]}}^\alpha$ is a centrality method, and the K-shell method is used here. $\sigma_{v_s^{[\alpha]} v_t^{[\alpha]}}^\alpha(v_x^{[\alpha]})$ denotes the number of shortest path entries of $v_s^{[\alpha]} \rightarrow v_t^{[\alpha]}$ through node $v_x^{[\alpha]}$ in layer α , and $\sigma_{v_s^{[\alpha]} v_t^{[\alpha]}}^\alpha$ denotes the number of shortest path entries of $v_s^{[\alpha]} \rightarrow v_t^{[\alpha]}$ in layer α .

Definition 7 (Combined Importance of Nodes): MLC is a combination of the above two aspects of interlayer influence and intralayer node importance, using the importance scores of auxiliary layer nodes passed along the interlayer links and then aggregating them with the importance scores of the core layer nodes themselves using internode transfer characteristics. Based on Definitions 1 and 2, $MLC(v_i)$ of node v_i can be defined as

$$MLC(v_i) = \sum_{\alpha=1}^N \omega_\beta^\alpha \times \sum_{(v_i^{[\rho]}, v_x^{[\alpha]}) \in E^{[\alpha, \beta]}} ILC_{v_x^{[\alpha]}}^\alpha \quad (8)$$

where N is the total number of layers of the multilayer heterogeneous network and $v_x^{[\alpha]}$ is the neighbor nodes of v_i within the α th layer.

Algorithm 1 provides the algorithmic description of MLC.

1) **Computational Complexity of MLC:** The method has two main components. In the first step: analyzing the importance of nodes of different degrees in the network from the perspective of degree metrics, we use the K-shell method to recursively strip the network nodes with time complexity of $O(n^2)$. In addition, we use betweenness centrality to measure the strength of control and constraint that the middle members in the network path have on the members at the ends of the path with a complexity of $O(n^2)$. In the second step, the time complexity of computing the interlayer influence in the network is $O(|N| \times |E|)$, where $|N|$ denotes the average

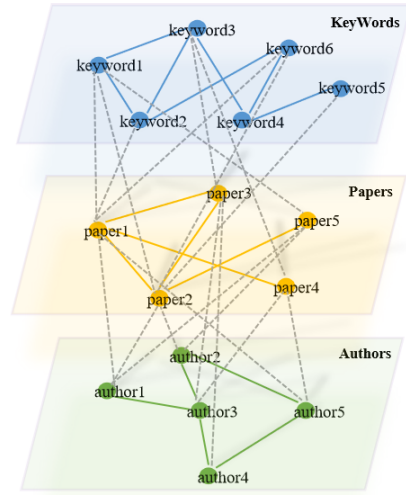


Fig. 2. Example of a three-layer heterogeneous citation network.

number of common neighbors of neighboring nodes in the interlayer network and $|E|$ denotes the size of the set E of edge connections, i.e., the time complexity is mainly influenced by the number of edge connections and the average number of common neighbors in the interlayer network. Therefore, the time complexity of MLC is $O(n^2 + |N| \times |E|)$.

2) **Explanation of MLC:** In this section, we use a simple multilayer heterogeneous citation network to explain the computational process of MLC. First, a citation network with a three-layer structure needs to be constructed. After the network is constructed, a three-layer network model with KeyWords, Papers, and Authors is obtained (see Fig. 2). We refer to the Papers layer as the core layer and similarly refer to the KeyWords and Authors layers as the auxiliary layers.

MLC focuses on the analysis of the core layer node importance, and thus, we choose to use the target node paper5 of the Papers layer to illustrate MLC.

First, we find all neighboring nodes of node paper5 in each layer and calculate the intralayer centrality scores of these nodes separately. Taking paper5's neighbor node keyword1 in the KeyWords layer as an example, there are

$$ILC_{keyword1}^{KeyWords} = \sum_{v_x \neq keyword1 \neq v_t} \frac{\sigma_{v_x v_t}^{KeyWords}(keyword1)}{\sigma_{v_x v_t}^{KeyWords}} + \phi_{keyword1}^{KeyWords} \quad (9)$$

where $ILC_{keyword1}^{KeyWords}$ denotes the intralayer centrality score of node keyword1. Since the importance of different types of relationship information varies for each node in the process of aggregation, interlayer influence weights are added in this article in order to reflect the different effects of various types of relationships on nodes. The process of calculating the interlayer influence weight is to give the weight of the auxiliary layer neighbors of node paper5. Taking keyword1 as an example by Definition 1, we can get $\omega_{paper5}^{KeyWords}$. In turn, we can extend the case of adding the interlayer influence weight $\omega_{paper5}^{KeyWords}$ to the above equation to the whole network,

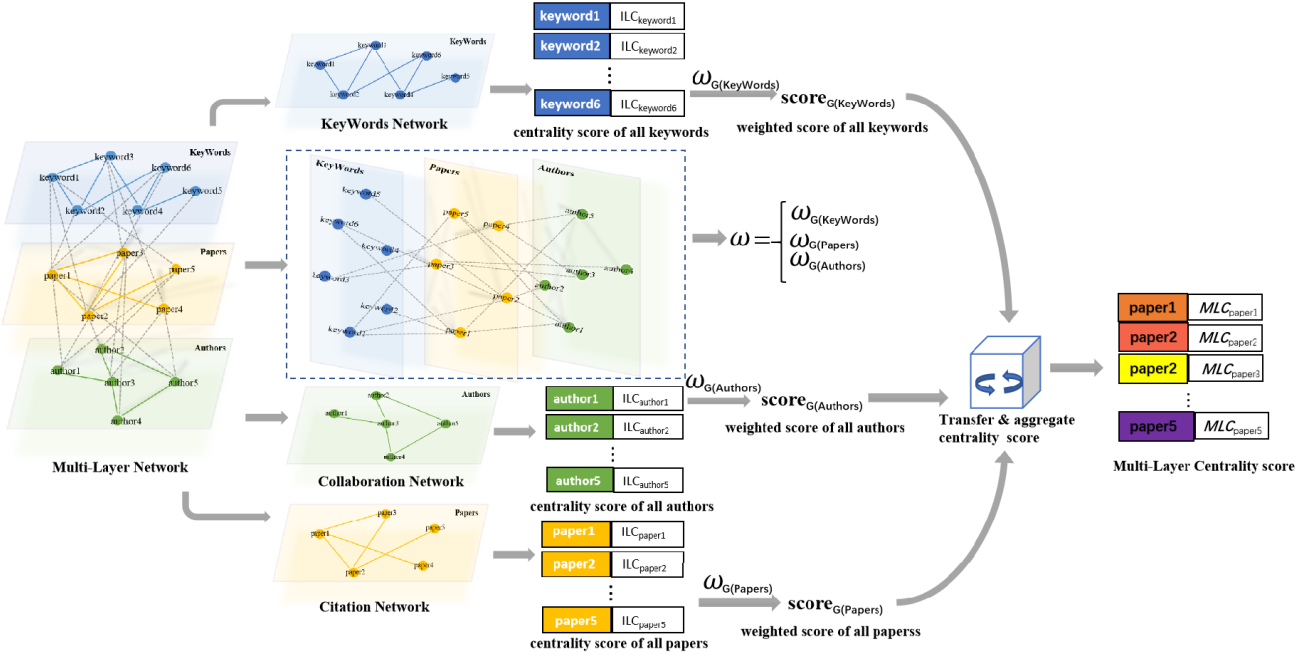


Fig. 3. Computational framework of MLC under three-layer heterogeneous citation network.

and then, we have

$$\text{MLC}(\text{paper5}) = \sum_{(\text{paper5}, v_x^{[\alpha]}) \in E^{[\alpha, \text{Papers}]}} \text{ILC}_{v_x}^{\alpha} \times \sum_{\alpha=\text{KeyWords}}^N \omega_{\text{Papers}}^{\alpha} \quad (10)$$

where N corresponds to each layer in the network and $\text{MLC}(\text{paper5})$ is the final score of node paper5 after aggregating the centrality of neighboring nodes at each layer. The overall framework of MLC proposed in this article is shown in Fig. 3. First, the importance of nodes within layers is analyzed layer by layer, and the weighted scores of nodes within each layer are obtained by combining the interlayer influence weights. Then, the importance scores of nodes within the auxiliary layer are passed along the interlayer links, and the importance scores of nodes within core layers are aggregated using the score transfer property between neighboring nodes. Finally, the final importance score sequence of nodes within the core layer is obtained.

IV. EXPERIMENTAL ANALYSIS

In this article, we propose a multilayer heterogeneous network node importance identification method incorporating MLC and validate the effectiveness of the method on several real network datasets. To ensure the reproducibility of the experiments, the information of the PC platform used for the experiments is given here. The experiments are conducted on a Dell PC, model DESKTOP-AGLQ8R5. The CPU model of this PC is Intel(R) Core(TM) i7-6700HQ with 8 G of RAM, and the system used is Windows 10 64bit. The IDE for the experiments is PyCharm.

A. Datasets

In order to verify the computational effectiveness of MLC, five multilayer heterogeneous citation networks extracted from the DBLP citation dataset (<https://www.aminer.cn/citation>) are used in the experiments to verify the effectiveness of the proposed method, which is described below in terms of dataset size and data characteristics.

First, we selected data from the DBLP citation network dataset for papers published between 2000 and 2019. Subsequently, we construct a multilayer heterogeneous citation network for every four years from 2000 to 2019, select the 100 most frequently cited papers as seed papers, and extract the cited paper data of the seed papers and the data of the papers citing them. Each network is an intricate multilayer heterogeneous citation network formed by multiple relationships among keywords, papers, and authors. The network is divided into three hierarchical structures, which are the cword network layer, the paper citation network layer, and the scholar-collaboration network layer. In each multilayer heterogeneous citation network, the cword network treats keywords as nodes and co-occurrence relations between keywords as edges, the paper citation network treats papers as nodes and citation relations between papers as edges, and the scholar cooperation network treats scholars as nodes and cooperation relations between scholars as edges. In addition, heterogeneous relationships between layers are also added to the network in consideration of maximizing the integration of as many node characteristics as possible when analyzing the nodes' influence. The connection between the cword network layer and the paper citation network layer is established by the keyword–literature affiliation, and the connection between the paper citation network layer and the scholar-collaboration network layer is established by the literature–scholar

TABLE I
MULTILAYER HETEROGENEOUS CITATION NETWORK DATA SCALE

Years		DBLP2000-2003	DBLP2004-2007	DBLP2008-2011	DBLP2012-2015	DBLP2016-2019
KeyWords	Node	4492	4879	5970	3851	2891
	Edge	17116	19235	24506	15167	10438
	$\max(d)$	82	79	96	79	81
	$\bar{d}$	7	7	8	7	7
	ρ	0.001697	0.001616	0.001375	0.002046	0.002499
	Clustering	0.8105	0.8076	0.8026	0.8003	0.8195
Inter-layer Links	Edge	8975	10145	12815	7915	5380
Papers	Node	1795	2029	2563	1583	1076
	Edge	7265	6410	8773	6262	4686
	$\max(d)$	412	236	164	298	807
	$\bar{d}$	8	6	6	7	8
	ρ	0.004512	0.003116	0.002672	0.005001	0.008102
	Clustering	0.4133	0.3815	0.3095	0.4648	0.4783
Inter-layer connection	Edge	5659	6704	8971	6098	4929
Authors	Node	3771	4619	5940	4082	4133
	Edge	7509	10403	12922	12258	19522
	$\max(d)$	54	59	60	87	184
	$\bar{d}$	3	4	4	6	9
	ρ	0.001056	0.000975	0.000733	0.001472	0.002286
	Clustering	0.7449	0.7717	0.8068	0.8449	0.9051

affiliation. After constructing each network, we took its maximum connected subgraph for each network as its multilayer heterogeneous citation network. Ultimately, these five network compositions from 2000 to 2019 are completed. The number of nodes and edges in the five networks is shown in Table I.

B. Evaluation Metrics Criteria

For identifying important nodes, the results of ranking the importance of nodes are usually evaluated using a transmission dynamics model, and in this article, the SIR model is used [53]–[55]. The classical SIR epidemic model includes the following three states: 1) susceptible state $S(t)$ indicates the number of nodes in a healthy state, susceptible to infection by others; 2) infection state $I(t)$ indicates the number of nodes that have been infected and can spread the disease to other individuals; and 3) recovery state $R(t)$ indicates the number of nodes that have recovered their health and will not be reinfected. In the SIR model, the infected node is the source of infection, and it transmits the infectious disease to its neighboring susceptible nodes through a certain probability β . Then, each infected node also has a certain probability μ that can be cured and immune to enter the recovery state; once the susceptible population is infected, it becomes a new source of infection again. The SIR model can be further expressed by using the differential equation as follows:

$$\begin{cases} \frac{dS(t)}{dt} = -\beta S(t)I(t) \\ \frac{dI(t)}{dt} = \beta S(t)I(t) - \mu I(t) \\ \frac{dR(t)}{dt} = \mu I(t) \end{cases} \quad (11)$$

where β denotes the infection probability and μ denotes the recovery probability.

In the SIR model, the evaluation index of node importance is mainly the transmission rate. In practical applications, there

are two ways to evaluate the importance of nodes according to the transmission rate index: one is timing evaluation, i.e., after the initial transmission source node is selected, the number of transmission time iteration steps is specified, and the transmission rates of different initial transmission source nodes within the same time iteration steps are compared. The other is quantitative evaluation, which means that after the initial transmission source node is selected, the experiment is stopped when the network transmission rate reaches a steady state, the time used to compare different initial transmission source nodes to reach the corresponding transmission rate is compared, and the shorter the time used indicates that the node is more important. For the sake of illustration, the transmission rate at the t th time iteration step is denoted as $r(t)$ and the transmission rate when the steady state is reached is denoted as $r(ts)$. Since the transmission process is probabilistically dependent on the transmission of infection, even if the experimental conditions of the two groups are identical, the final results obtained will be somewhat different. Therefore, the expected value of the results obtained from 100 independent experiments is chosen as the evaluation result.

C. Effectiveness

In this section, five different sets of experiments are conducted on five datasets and evaluated by the SIR model to verify the accuracy of the proposed algorithm in measuring the importance of nodes in terms of the transmission range and transmission rate of each measure.

Experiment 1: The accuracy of each method to measure the importance of nodes is evaluated by observing the transmission range and transmission rate of SIR transmission curves. In each network, the nodes are sorted in descending order according to the measure of node importance by each method, and the top 1% of nodes is selected as the initial infectious source nodes, and the transmission rate $r(t)$ for a given number of time iteration steps is counted and the SIR transmission curve is plotted, where the horizontal axis

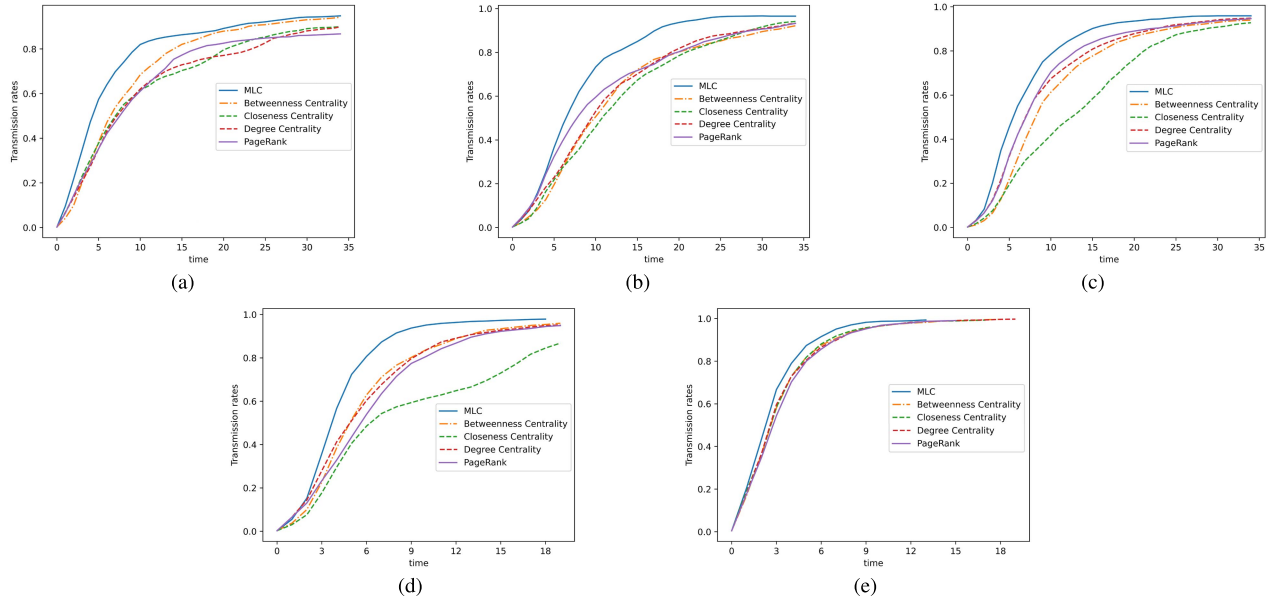


Fig. 4. Comparison of the growth of the propagation rate of each important node identification method over time. (a) DBLP2000-2003. (b) DBLP2004-2007. (c) DBLP2008-2011. (d) DBLP2012-2015. (e) DBLP2016-2019.

TABLE II
TRANSMISSION RATE OF IMPORTANT NODE IDENTIFICATION METHODS IN A STEADY STATE

Years	DBLP2000-2003	DBLP2004-2007	DBLP2008-2011	DBLP2012-2015	DBLP2016-2019
Degree Centrality	0.906	0.937	0.95	0.956	0.995
PageRank	0.868	0.941	0.947	0.953	0.988
Betweenness Centrality	0.943	0.928	0.942	0.964	0.997
Closeness Centrality	0.901	0.947	0.935	0.881	0.993
MLC	0.949	0.966	0.959	0.979	0.994

represents the number of transmission time iteration steps. The vertical axis represents the corresponding transmission rate $r(t)$, and the transmission rate of the source node is illustrated by observing the time when the curve reaches the steady state and the corresponding transmission rate $r(ts)$. The experimental results shown in Fig. 4 indicate that the following conditions hold.

On the five networks, MLC has improved the transmission rate compared with the initial transmission nodes selected by other methods, and the performance results are better than other methods. The results of the closeness centrality evaluation are less than those of MLC and other centrality measures, especially on the DBLP2008-2011 and DBLP2012-2015 datasets. The overall performance of the betweenness centrality is excellent, but it is inferior to the proposed MLC. The difference between these two methods is especially obvious in the first four networks.

Experiment 2: To further discuss the correlation between the measured values of the nodes of each method and their transmission capacity and to evaluate the accuracy of each method in measuring the importance of the nodes. This experiment ranks the steady-state values of $r(ts)$ obtained for each node in the SIR model. Table II depicts the transmission ability of each node in the SIR model in the five networks. The horizontal axis in Fig. 5 indicates the measured value of each method under the corresponding dataset, and the vertical

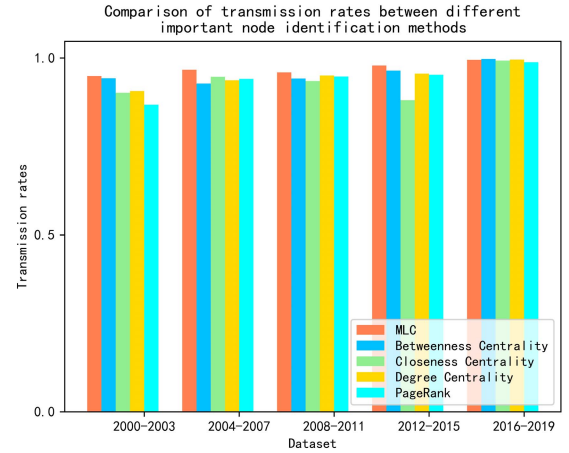


Fig. 5. Transmission rates of the transmission source nodes in the steady state for each method.

axis indicates the steady-state value $r(ts)$ obtained by each method in the SIR model.

To more clearly show the transmission rate enhancement effect of MLC, Table II and Fig. 5 show the transmission rates of the transmission source nodes in a steady state for each method. From the analysis of the results, the transmission effect of the transmission source nodes obtained by MLC is

significantly better than the other methods; in contrast, the results obtained by the closeness centrality method are poorer.

V. CONCLUSION

In this article, we propose a multilayer heterogeneous network node importance identification method incorporating MLC. It is studied from the perspective of multiple types of relationships and network structure features, divides heterogeneous nodes into different layers (core layer and auxiliary layer), further considers the layer influence weights of the core layer and auxiliary layer, and finally expresses the importance of core layer nodes by fusing the centrality score of auxiliary layer nodes and the centrality score of core layer nodes. It not only simplifies the heterogeneous network structure and preserves multiple relationship information but also makes full use of the network structure information and improves the performance of identifying important nodes. It is validated on several real multilayer heterogeneous citation datasets and its efficiency is evaluated using the SIR model. The experimental results based on these real networks validate the effectiveness of the method and demonstrate that the performance of MLC outperforms existing methods. However, MLC is limited to unweighted and undirected network scenarios. To address these challenges, future research will extend MLC to weighted undirected, weighted directed, and other networks to accommodate more heterogeneous network analysis scenarios.

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Liangtian Wan (Member, IEEE) received the B.S. and Ph.D. degrees from the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China, in 2011 and 2015, respectively.

From October 2015 to April 2017, he was a Research Fellow with the School of Electrical and Electrical Engineering, Nanyang Technological University, Singapore. He is currently an Associate Professor with the School of Software, Dalian University of Technology, Dalian, China. His current

research interests include graph learning, data science, and array signal processing.

Dr. Wan has been serving as an Associate Editor for the IEEE ACCESS AND *Journal of Information Processing Systems*.



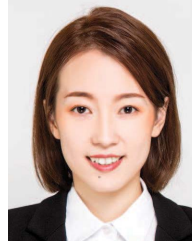
Mingyue Zhang received the B.S. degree from the College of Information Science and Engineering, Xinjiang University, Xinjiang, China, in 2019. She is currently pursuing the master's degree with the School of Software, Dalian University of Technology, Dalian, China.

Her research interests include network science and graph learning.



Xiaona Li received the bachelor's degree from the School of Information Science and Technology, Dalian Maritime University, Dalian, China, in 2016. She is currently pursuing the master's degree with the School of Software, Dalian University of Technology, Dalian.

Her current research interests include graph learning and graph signal processing.



Lu Sun received the B.S. and Ph.D. degrees from the School of Software, Dalian University of Technology, Dalian, China, in 2013 and 2019, respectively.

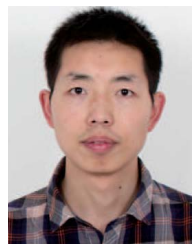
She is currently an Assistant Professor with the Institute of Information Science Technology, Dalian Maritime University, Dalian. Her current research interests include offshore communication, resource allocation, and wireless power transfer.



Xianpeng Wang (Member, IEEE) was born in 1986. He received the M.S. and Ph.D. degrees from the College of Automation, Harbin Engineering University (HEU), Harbin, China, in 2012 and 2015, respectively.

He was a full-time Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2015 to 2016. He is currently a Professor with the College of Information and Communication Engineering, Hainan University, Haikou, China.

He is the author of more than 60 articles published in related journals and international conference proceedings and was a reviewer of more than 20 journals. His major research interests include communication systems, array signal processing, radar signal processing, compressed sensing, and its applications.



Kaihui Liu (Member, IEEE) received the Ph.D. degree in communications and information systems from the National Key Laboratory of Science and Technology on Communications, University of Electronic Science and Technology of China, Chengdu, China, in 2019.

He is currently a Research Engineer with the University of Electronic Science and Technology of China. His research interests are in the mathematics of data science, particularly the interplay of convex and nonconvex optimization, statistical learning,

high dimensional statistics, signal processing, computational imaging, and wireless communication.